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RESEARCH ARTICLE

Increased habitat fragmentation leads to isolation among and low genetic diversity within populations of the imperiled Kentucky Arrow Darter (*Etheostoma sagitta spilotum*)

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Abstract

The Kentucky Arrow Darter (*Etheostoma sagitta spilotum*) is federally listed as threatened and known only from five tributary systems in the Kentucky River system, Kentucky, USA. Recent surveys revealed considerable population decline, with individuals detected in only 45% of known historical localities. Impacts of these declines on the genetic structure of *E. s. spilotum* may be exacerbated by limited dispersal capabilities and small initial population size resulting in high levels of isolation among extant populations. Long standing genetic isolation may also be evident in contemporary genetic signatures; populations that have undergone historic isolation may be prime candidates for intensive conservation efforts. To address contemporary and historic genetic isolation, we generated genotypic (11 microsatellite loci) and *mt*DNA sequence (ND2 gene) data from multiple locations spanning the taxon's range. We recovered seven haplotypes with low divergence levels, shared among and within multiple Kentucky River tributary systems, indicating absence of long-standing isolation among localities examined. In contrast, microsatellite data suggested all nine populations are functionally isolated, with little to no admixture among populations, even among those within the same tributary system. The drastic decline of *E. s. spilotum* populations has likely combined with limited dispersal, resulting in extensive contemporary genetic isolation among extant populations. Conservation management plans to enhance stability and maintain survivability of *E. s. spilotum* must address the severe genetic isolation identified here and work towards increasing gene flow among extant populations.

Keywords Fish · Allelic diversity · Effective population size · Population structure · Microsatellites · Phylogeography

Introduction

For many species, streams are heterogeneous landscapes, and optimal habitats are patchily distributed in linear or dendritic arrangements (Gotelli and Taylor 1999; Fagan 2002; Fagan et al. 2002; Kottelat and Freyhof 2007). Artificially increased patchiness due to habitat fragmentation can lead to reduced population connectivity and is intimately linked to declines in population persistence, recolonization potential, and global biodiversity (Young and Clarke 2000; Kottelat and Freyhof 2007). Habitat fragmentation is a widely recognized conservation concern for riverine fishes of the Southeastern U.S. and worldwide (Warren et al. 2000; Jelks et al. 2008). Reductions in population size and connectivity associated with habitat fragmentation often lead to rapid losses of genetic diversity through genetic drift and inbreeding. Because genetic diversity is one key attribute linked to population persistence in changing environments, loss of genetic diversity may cause reduced adaptive potential and ultimately result in population extinction (Reed and Frankham 2003).

The magnitude of response of fishes to habitat fragmentation varies and may be largely species-specific or context-specific (e.g., reviewed by Ewers and Didham 2006; Roberts

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et al. 2013; Amos et al. 2014; Ginson et al. 2015). However, small-bodied, benthic fishes, such as darters (Percidae: Etheostomatinae) seem particularly susceptible to the effects of habitat fragmentation. For example, Roberts et al. (2013) found that increased habitat fragmentation caused by dams led to genetic fixation among populations of Roanoke Logperch (*Percina rex*) equivalent to that seen among populations separated by 1200 km. Sterling et al. (2012) found that channelized stream reaches were barriers to dispersal and gene flow in the Yazoo Darter (*Etheostoma raneyi*). Most darters possess a suite of adaptive traits, including lack of a functional swimbladder (an adaptation to a benthic lifestyle), a small body, and high reproductive investment (Page 1983). These characteristics reduce dispersal potential and are associated with species susceptible to habitat fragmentation (Turner et al. 1996; Turner and Trexler 1998; Turner 2001).

The Kentucky Arrow Darter, *Etheostoma sagitta spilotum,* is a headwater-stream (first- to third-order) specialist restricted to five tributary systems in the upper Kentucky River system of eastern Kentucky [Fig. 1a and Online Resource 1; Kuehne and Barbour 1983; Burr and Warren 1986; US Fish and Wildlife Service (USFWS) 2013]. In 2005, it was identified as a "species of greatest conservation need" by the Kentucky Department of Fish and Wildlife Resources (KDFWR 2005), and in 2010, as a candidate for federal listing under the Endangered Species Act (ESA; USFWS 2010). Recent surveys (conducted between 2007 and 2012) by the US Fish and Wildlife Service (USFWS) and Kentucky Department of Fish and Wildlife Resources (KDFWR) found *E. s. spilotum* at only 45 of the 100 known historical localities (data from USFWS 2013; Online Resource 1). Declines in occurrence and abundance were attributed to degraded habitat conditions resulting from increased sedimentation and other non-point source pollutants stemming from a variety of anthropogenic activities, including surface mining, urbanization, logging, and agriculture (Thomas 2008; USFWS 2013). Hitt et al. (2016) found that reduced forest cover and increased instream conductivity have particularly negative impacts on persistence and abundance of *E. s. spilotum*. Analyses of historical localities showed that *E. s. spilotum* was rarely detected in streams with conductivity levels over 230 µS (Thomas 2008). The observed declines in both abundance and range, combined with the occurrence of multiple threats to existing habitat, prompted the USFWS to list the Kentucky Arrow Darter as a threatened species under the Endangered Species Act (USFWS 2015).

Loss of *E. s. spilotum* from nearly 50% of its former range and the continued degradation of corridors connecting remaining populations has resulted in a highly fragmented distribution (Online Resource 1; Thomas 2008; USFWS 2013). Like many other darters, *E. s. spilotum* has a suite of adaptive traits (Page 1983) that reduce its dispersal potential and are associated with species more susceptible to habitat fragmentation (Turner et al. 1996; Turner and Trexler 1998; Turner 2001). Additionally, *E. s. spilotum* maintains low population densities (Thomas 2008; USFWS 2013; Baxter 2015 unpubl. M.S. thesis), requires specific spawning habitat, and exhibits territoriality among males during the spawning season (Kuehne and Barbour 1983). Given these attributes, contemporary *E. s. spilotum* effective populations may be small and isolated by distances greater than those typically traversed by adults or larvae. Small, isolated populations are more likely to suffer loss of genetic diversity through genetic drift or other stochastic events that reduce a population's adaptive potential (Frankham 1998; Frankham et al. 1999; Spielman et al. 2004; Frankham 2005). Alternatively, this same suite of adaptive traits suggests potential for long-standing isolation among populations stemming from natural instream barriers or filters to dispersal. For example, movements of headwater-stream specialists, such as *E. s. spilotum*, may be restricted by water depth, changes in stream gradient and substrate composition, or past vicariant events. This can result in high levels of genetic structure among populations, particularly among those in different tributary systems (Hollinsgworth and Near 2009; Fluker et al. 2010). In some cases, historical events can have a more pronounced influence than contemporary events on genetic structure and diversity. Thus, both should be considered when evaluating spatial genetic diversity as a basis for conservation management actions for imperiled species (Davis et al. 2015).

The impacts of recent habitat fragmentation on genetic diversity relative to potential long-standing geographic isolation of *E. s. spilotum* populations are unknown. Thus, our objectives were to examine contemporary demographic and historical patterns of genetic diversity and structure across the range of *E. s. spilotum* using microsatellite loci and the mitochondrial ND2 gene, respectively. Mitochondrial DNA is commonly used to identify historical levels of genetic structure and can provide a basis for conservation of long-standing, isolated lineages; this structure also provides a baseline for historical levels of genetic exchange (Morales et al. 2011; Blakney et al. 2014). In contrast, microsatellites are commonly used to infer contemporary patterns of gene flow and genetic structure (McCraney et al. 2010; Blakney et al. 2014; Sterling et al. 2012). Thus, use of these two types of markers can help distinguish long-standing genetic structure from that due to recent losses in connectivity among populations (Chiucchi and Gibbs 2010; Blakney et al. 2014; Davis et al. 2015).

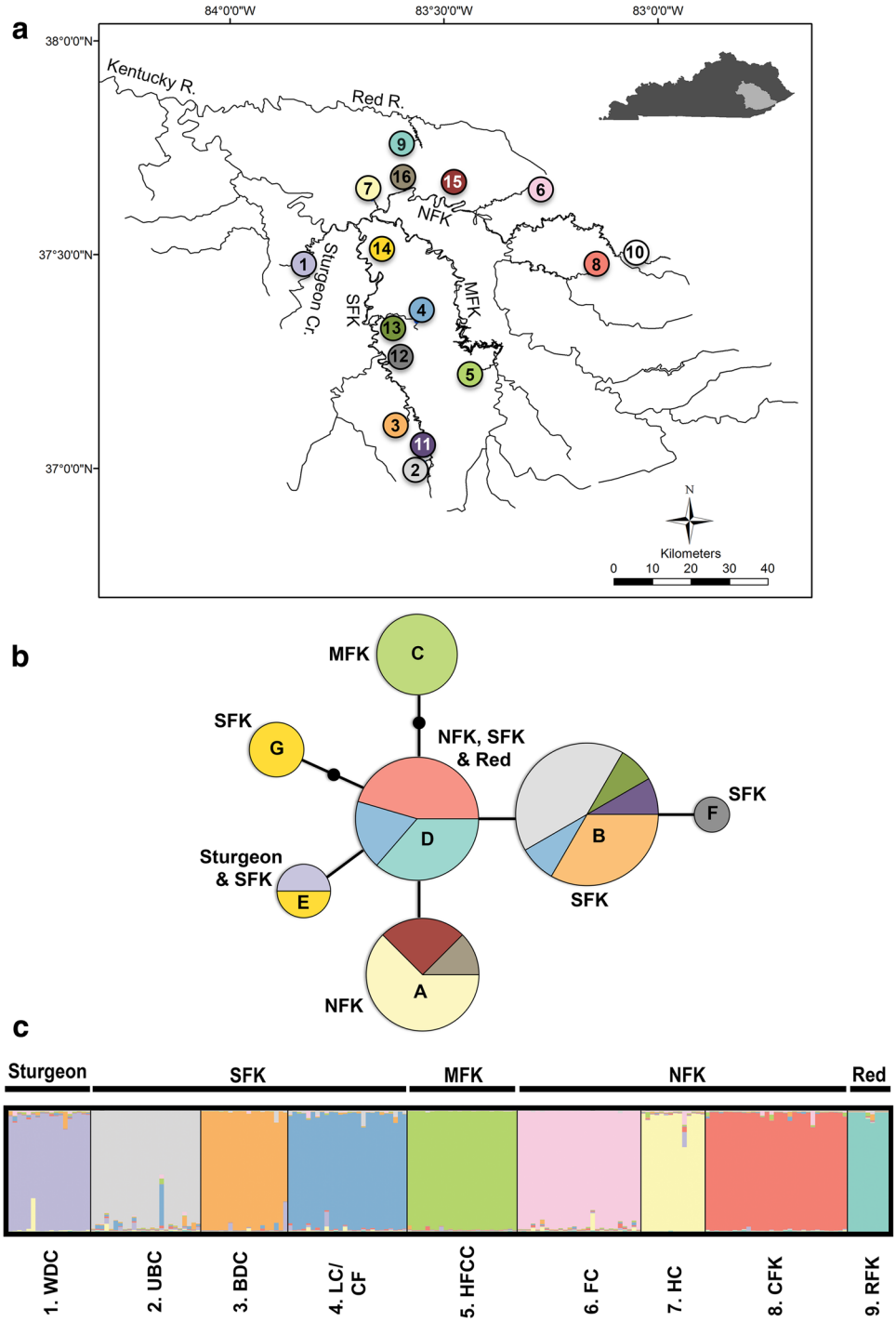

Fig. 1 **a** Localities sampled for assessment of genetic diversity in the Kentucky Arrow Darter, *Etheostoma s. spilotum*. Colors and numbers denote locality and population identifiers used in the text; specific locality information provided in Supplemental Materials (Online Resource 2). Mitochondrial DNA sequences were generated for sites 1–5, 7–9, and 11–16; microsatellite genotypes for sites 1–10. **b** Resulting haplotype network of mitochondrial ND2 sequences recovered for *E. s. spilotum*. Letters denote different haplotypes. Size of pies reflect number of individuals with a given haplotype and colors correspond to those in 1a to denote locality of occurrence for a given haplotype. **c** Results of the STRUCTURE analysis of 11 microsatellite loci showing K = 9. Boxes designate populations recovered and color of bars within each box denote population of ancestry of each individual, corresponding to those in 1a and b. Bars and labels above denote system of origin for each population. Numbers below correspond to population identifiers of 1a and used in text: 1. WCD = Wild Dog Creek; 2. UBC = Upper Bear Creek; 3. BDC = Big Double Creek; 4. LC/CF = Laurel Creek/Cortland Fork; 5. HFCC = Hell for Certain Creek; 6. FC = Fighting Creek; HC = 7. Hell Creek; 8. CFK = Clemons Fork; 9. RFK = Rockbridge Fork

Methods

Localities examined

Etheostoma sagitta spilotum was collected using standard seining and electrofishing techniques, and fin clips were taken from each individual and stored in 95% ethanol. Samples were collected throughout the known range of the species; individuals from multiple sites within a river system were examined, where possible, to assess both within and among tributary system variation (Fig. 1a and Online Resource 2). Tissues used to generate the mtDNA sequences were taken from a total of 40 individuals from 14 localities (Fig. 1a; sites 1–5, 7–9, 11–16). Samples for these analyses were collected and sequenced in 2008 and represented all streams sampled for the microsatellite data analysis, except

Quicksand Creek (NFK) and Frozen Creek (NFK); for the latter, a proximate tributary to the NFK, Holly Creek, was examined instead.

Tissues for microsatellite data generation were taken from individuals collected in 2013–2014 from 12 localities. We grouped samples from Cortland Fork and Laurel Fork as a single population a priori because they were part of the same watershed and only ~3 km apart. Similarly, two localities in Hell Creek, a North Fork Kentucky River (NFK) tributary, were considered a single population. Thus, a total of ten putative populations were examined (Fig. 1a; sites 1–10).

Mitochondrial DNA collection and analysis

We inferred levels of historical genetic (or phylogeographic) structure using the mitochondrial ND2 gene. Whole genomic DNA was extracted from individuals using a 5% chelex solution with 2 µl of Proteinase K and overnight digestion. Two microliters of template DNA and previously published primers (Kocher et al. 1995) were used in a 25 µl volume PCR reaction using a touchdown cycling protocol: initial denaturation at 94 °C for 30 s, followed by 5 cycles each of denaturation at 94 °C for 30 s, annealing at 57–55 °C for 30 s, and extension at 72 °C for 90 s, followed by 20 cycles of 94 °C for 30 s, 54 °C for 30 s, and 72 °C for 90 s and a final extension at 72 °C for 10 min. Resulting PCR products were purified using ExoSap-It (USB) prior to sequencing. Purified PCR products were sequenced using Sanger methods by the UF-ICBR Sanger Sequencing Facility. Resulting sequences were aligned and edited using CodonCode Aligner (v.1.6, Soft Genetics).

To examine geographic variation and relationships among resulting ND2 haplotypes, haplotype networks were constructed using a statistical parsimony analysis (Templeton et al. 1992; Castelloe and Templeton 1994) with 95% limit imposed on connections (TCS v 1.2.1). Population equilibrium tests were conducted in DNASP v. 5.10.01 (Librado and Rozas 2009) using Tajima's D (Tajima 1989), Fu's Fs (Fu 1997) and R2-tests (Ramos-Onsins and Rozas 2002). For all tests, significance was assessed using a null distribution of 10,000 coalescent simulations.

Microsatellite genotyping

A set of 11 darter microsatellite loci was identified from the literature (Porter et al. 2002; Gabel et al. 2008; Switzer et al. 2008; Table 1) and optimized for amplification and allelic variation. Whole genomic DNA was extracted from preserved fin tissue using the Fisher GeneJet Genomic DNA Purification kit and diluted to 10 ng/µl for use in microsatellite genotyping reactions. Forward primers were modified with an M13 tag (CACGACGTTGTAAAACGAC) at the 5′

Table 1 Forward and reverse primers for 11 loci used in microsatellite genotyping of the Kentucky Arrow Darter, *Etheostoma s. spilotum*

Locus ID	Oligo sequence	Focal species	Source	ABI dye label
EosC2	F 5′-GCTCTCACAAACACACACAAAC-3′ R 5′-ATCGACTCAACCCCAGATTAG-3′	*E. osburni & E. variatum*	Switzer et al. (2008)	6Fam
EosC3	F 5′-CAGCATTTTCAGGTCATACCAT-3′ R 5′-GCTTTGGTTTCTCAGCTACTCC-3′	*E. osburni & E. variatum*	Switzer et al. (2008)	VIC
EosC6	F 5′-AAAGCCTGAGGGACAATTACAC-3′ R 5′-CCTTTGCTGGTAAATTCTACAC-3′	*E. osburni & E. variatum*	Switzer et al. (2008)	NED
EosC117	F 5′-CCTCACACACTGCTCAAAGAG-3′ R 5′-GGGAGTCATTTGGGTTCAGT-3′	*E. osburni & E. variatum*	Switzer et al. (2008)	PET
EosD10	F 5′-CCGGGCTGAAAGTCTTATG-3′ R 5′-ACACGTTCATGTCATTTGTTTC-3′	*E. osburni & E. variatum*	Switzer et al. (2008)	6Fam
EosD11	F 5′-ACCAGATGCAGTGGATGAATAT-3′ R 5′-GCGGTATCTAATGCTATTTCCC-3′	*E. osburni & E. variatum*	Switzer et al. (2008)	VIC
EosD107	F 5′-CATTTAACATTCCCTGGTTGTG-3′ R 5′-TTGCAGTGCAGTGGAGTTTTA-3′	*E. osburni & E. variatum*	Switzer et al. (2008)	NED
EosD116	F 5′- GCTGCCGACAGTGAAATAATAC-3′ R 5′-GTGCATGTTTGTTGTGTTATGG-3′	*E. osburni & E. variatum*	Switzer et al. (2008)	PET
EosD131	F 5′-AAAAAGGGGGACAGTGTGTC-3′ R 5′-GCATCAGCAAATAGGCAGAG-3′	*E. osburni & E. variatum*	Switzer et al. (2008)	6Fam
Esc26b	F 5′-CAATGCGCCACATTGAGAAGG-3′ R 5′-GCACAACATATGTCGTTAAGCTC-3′C	*E. scotti*	Gabel et al. (2008)	VIC
Cv12	F 5′-ACTGTTAGCCCTACACTCTG-3′ R 5′-TTGAAGCAGGTATTCTCACC-3′	*E. virgatum*	Porter et al. (2002)	NED

end, and PCR was conducted using the three-primer method (Boutin-Ganache et al. 2001). Each locus was assigned a dye-label (6-FAM, PET, VIC or NED), which was used throughout the project (Table 1). Loci were amplified in 10 µl reactions consisting of 1 X Taq Buffer, 3 mM MgCl2, 0.2 mM dNTPs, 0.25 µM forward primer, 0.5 µM reverse primer, 0.1 µM M13-labeled primer, and 0.5 U Taq Polymerase. PCR cycle conditions included an initial denaturation step of 94 °C for 1 min, followed by 35 cycles of 94 °C for 30 s, 60 °C for 30 s, and 72 °C for 30 s, with a final extension at 72 °C for 5 min. Genotypes were generated by the University of Florida Interdisciplinary Center for Biotechnology Research (UF-ICBR) Genotyping Core using an ABI 3730 sequencer. Control samples were included on each plate to ensure consistency in allele calls across runs. Genotypes were scored using GeneMarker v. 1.6.

Microsatellite data were assessed for evidence of errors in genotyping based on large allele dropout, stutter, and null alleles (1,000 simulations with 95% CI) using Microchecker v 2.2.3 (Oosterhout et al. 2004). To confirm loci independence, we assessed linkage disequilibrium (LD) using GenePop v. 4.4 (Rousset 2008) with default Markov Chain (MC) parameters (dememorization number set to 1000, 100 batches, and 1000 iterations per batch). Additionally, GenePop v. 4.4 was used to assess Hardy–Weinberg equilibrium for each population, with alpha determined via Bonferroni Correction for multiple tests (Cabin and Mitchell 2000). Pairwise fixation indices [F_{ST}, or Θ of Weir and Cockerham (1984)], inbreeding coefficient (F_{IS}) per population, and observed (H_o) and expected (H_e) heterozygosity were calculated in Genetix 4.05.2 (Belkhir et al. 1996). Significance of F_{ST} and F_{IS} was tested via 1000 permutations of the data. Allele frequencies for each locus within populations and Chi Squared tests for Hardy–Weinberg Equilibrium (HWE) per locus per population were calculated in GenAlEx 6.502 (Peakall and Smouse 2006, 2012).

We used a Bayesian clustering method implemented in STRUCTURE v 2.3.4 to infer the most likely number of populations (K) (Pritchard et al. 2000; Falush et al. 2003, 2007), assessed both by mean log-likelihood posterior probability and estimation of ΔK (Evanno et al. 2005). We implemented a genetic admixture model with correlated allele frequencies and 10 iterations for each possible K for 500,000 post-burnin replications (burn-in was 20,000 generations). Data were summarized and visualized using Structure Harvester (Earl and vonHoldt 2012) and imaging of STRUCTURE output data was facilitated by CLUMPP (Jakobsson and Rosenberg 2007) and DISTRUCT (Rosenberg 2004).

To assess whether genetic structure (F_{ST}) is correlated to geographic distance [river-kilometers, measured in ArcMap v.10.2.2 (ESRI, CA)], a Mantel test was conducted using Isolation By Distance Web Service v 3.24 (Jensen et al. 2005) with 10,000 randomizations of the data. River distances were log-transformed to decrease point spread around the best-fit line and to identify potential outliers. Reduced major axis regression was used to assess the strength of the correlation between genetic structure and geographic distance (Jensen et al. 2005).

Effective population size (N_e) was estimated using the LD method (Waples and Do 2008), which allows estimates of current N_e using single-sample datasets. The analysis was implemented in NeEstimator v2.01 (Do et al. 2014), with 0.05 set as the lowest allele frequency cut-off due to the small sample size of some of the populations (Waples and Do 2008).

BOTTLENECK v. 1.2.02 was used to identify recent, extreme population decline in focal populations (Cornuet and Luikart 1996); only those with more than ten individuals (as required by program assumptions) were included. We tested for heterozygote excess across populations using a two-phase model with 0, 10, and 20% multistep mutations (as recommended in Peery et al. 2012) and variance set at 36 (see Di Rienzo et al. 1994). Deviation from the null hypothesis of populations being in drift-mutation equilibrium was tested using a one-tailed Wilcoxon sign ranks test with 1000 replications (Luikart and Cornuet 1998).

Results

Mitochondrial DNA

Seven ND2 haplotypes were recovered from the 40 individuals sequenced. Haplotypes had low divergences (< 1.0%) and differed from one another by one to four mutations (Fig. 1b). The star-shaped pattern of the haplotype network implies historical population expansion (Rogers and Harpending 1992), which was supported by a significant and negative Tajima's D (−2.0907; p = 0.002). Other population equilibrium tests were not significant (Fu's F; p = 0.55) or were positive (R-2 = 0.000, p = 0.000), suggesting other demographic processes may have shaped the observed pattern. Haplotype D was inferred as the most ancestral haplotype sampled.

Of the 14 populations examined, all but two were comprised of a single haplotype (Fig. 2a). Only Laurel/Cortland Fork [population (pop) 4; SFK)] and Lower Buffalo Creek (pop 14; SFK) had two haplotypes. At the river system level, the SFK had the highest haplotypic variation (five haplotypes). Three sites had unique haplotypes (Lower Buffalo Creek—SFK, pop 14; Bullskin Creek—SFK, pop 12; and Hell for Certain Creek—MFK, pop 5; Fig. 2a). At the system level, unique haplotypes were found in the SFK [haplotype (hap) B, F, and G)], NFK (hap A), and MFK (hap C) rivers.

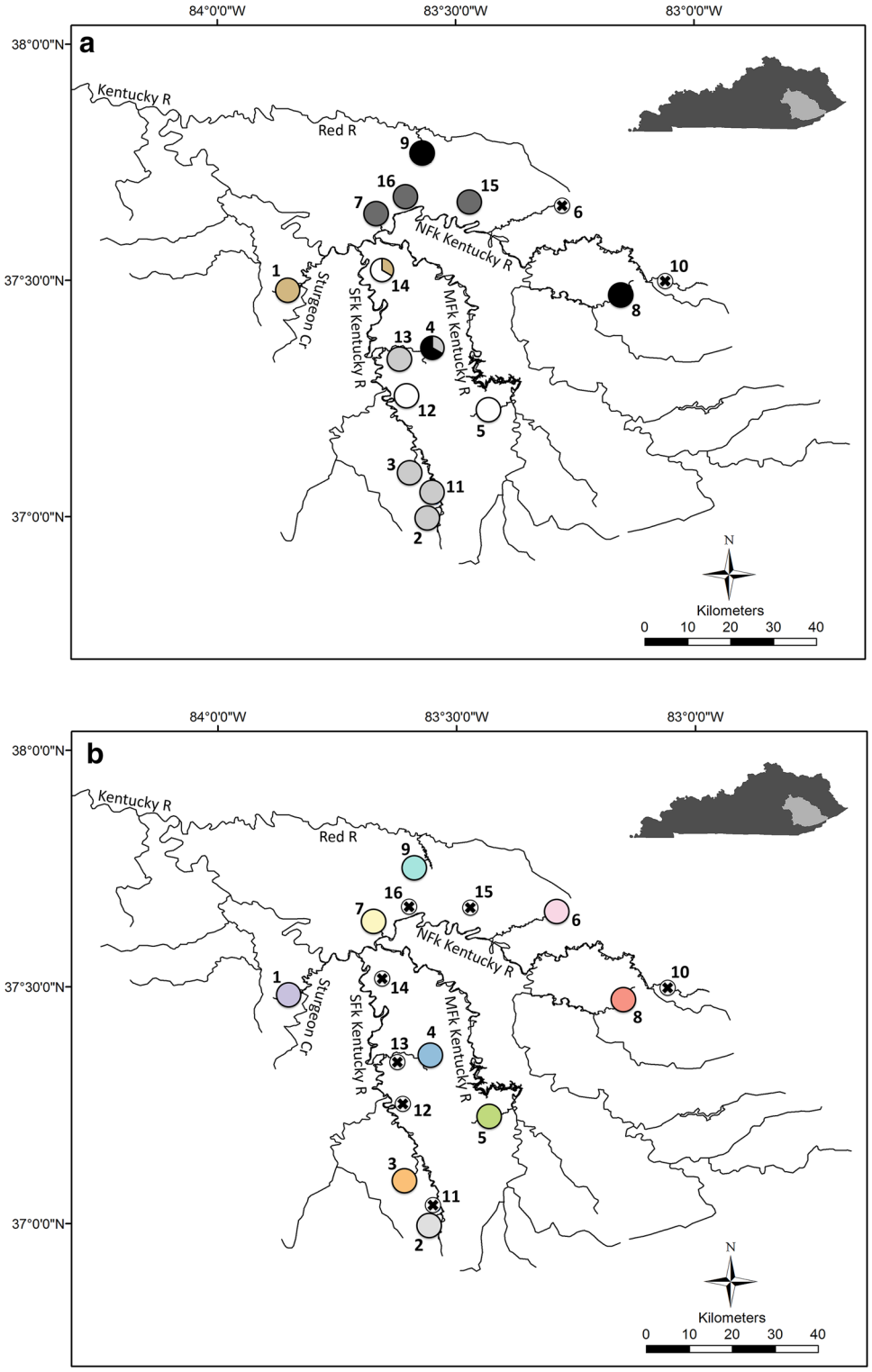

Fig. 2 **a** Sites examined using the mitochondrial ND2 gene with pie charts that display the proportion of unique (white) and shared haplotypes (colors) recovered among Kentucky Arrow Darter, *Etheostoma s. spilotum*, populations. **b** Geographic distribution of *E. s. spilotum* populations (K = 9; different colors correspond to genetically distinct populations) recovered from the STRUCTURE analysis of 11 microsatellite loci. In both, an "x" denotes a site not examined for that data set

Several sites in the SFK shared haplotype B (pops 2, 3, 11, 13, 14) and haplotype A was shared among multiple localities within the NFK (pops 7, 15, and 16). Two haplotypes had distributions spanning river system divides: haplotype D was the most broadly distributed, shared among populations (pops 4, 8, and 9) in the Red,

SFK, and NFK river systems and haplotype E was shared between populations (pops 1 and 14) from the SFK and Sturgeon Creek (Fig. 2a).

Microsatellites

We genotyped 213 individuals for 11 microsatellite loci from the 10 a priori designated populations from which microsatellites were examined (Fig. 1a). Missing data due to low DNA yield for specimens from Middle Fork Quicksand Creek (pop 10, NFK) precluded inclusion of this population in analyses. There was no evidence of errors in the data due to large allele dropout, stutter, or null alleles, and all alleles were polymorphic. Thus, all loci were used in subsequent analyses.

Nine to 31 individuals were amplified per population (Table 2). Allele richness ranged from 2 to 28 alleles across the 11 loci and all populations (mean alleles per locus = 10.18). Many, particularly Rockbridge Fork (pop 9; Red River), Big Double Creek (pop 3; SFK) and Hell for Certain Creek (pop 5; MFK), had a low percentage of polymorphic loci and a relatively high percentage of fixed alleles. Within populations, the average number of alleles per locus was low (range 1.27–5.6; Table 2). Three populations were not in HWE, indicating homozygote excess; after Bonferroni correction, only Clemons Fork (pop 8; NFK), showed significant deviation from HWE (Table 2). Although some analyses assume HWE for all populations, significant deviation from HWE is likely for species or populations that have experienced significant declines or range reductions (Hartl and Clark 1997, Wigginton et al. 2005), such as that observed for *E. s. spilotum*.

Metrics of heterozygosity and genetic fixation suggest a high degree of genetic isolation (Tables 2 and 3). Variation in mean observed (H_o) and expected (H_e) heterozygosity was high among populations (Table 2). Rockbridge Fork (pop 9; Red River) had the lowest H_o ($H_o = 0.020$) and Wild Dog Creek (pop 1; Sturgeon Cr.) had the highest ($H_o = 0.630$), and the overall average H_o was 0.426, lower than averages observed for several other imperiled darter species (Table 4). Genetic fixation (F_{ST}) was high (0.3605) for all populations and all pairwise comparisons were significant ($p \leq 0.0001$; Table 3). Big Double Creek (pop 3; SFK), Rockbridge Fork (pop 9; Red River) and Hell for Certain Creek (pop 5; MFK) had some of the highest pairwise F_{ST} values, suggesting they are among the most genetically isolated populations in the dataset. Overall inbreeding coefficient (F_{IS},) was 0.0252 (95% CI −0.0218–0.0826), and most of the population-level F_{IS} values trended higher than the expected value of 0 (Table 2); however, all 95% confidence intervals overlapped 0. Therefore, we cannot confidently assess the level of inbreeding within populations.

The STRUCTURE analyses recovered nine distinct clusters (K = 9), using both mean log-likelihood and ΔK approaches (Fig. 1c and Online Resource 3 for likelihood and ΔK graphs); clusters corresponded to each locality or a priori grouping examined. The two a priori population

Table 2 Mean number of individuals genotyped (n), mean number of alleles per locus (N_A), average observed (H_o) and expected (H_e) heterozygosity, inbreeding coefficient (F_{is}), percentage (%) of polymorphic loci, p-values from Hardy–Weinberg equilibrium (HWE) exact tests (significant values in bold; **denotes significance after Bonferroni correction), and estimates of effective population size (N_e; *denotes upper bound estimates below 200) for the nine populations of Kentucky Arrow Darter, *Etheostoma s. spilotum*, examined using microsatellite data

Populations by River System	n	N_A	H_o	H_e	F_{IS} (95% CI)	HWE	% polymorphic loci	N_e (95% CI)
Sturgeon Creek								
1. Wild Dog Creek	18	5.63	0.63	0.607	-0.0088 (− 0.1381 to 0.05386)	**0.0163**	100%	42.3 (19–1227)
South Fork Kentucky R.								
2. Upper Bear Creek	24	5.45	0.598	0.583	-0.0033 (− 0.09624 to 0.04810)	0.1045	91%	36.6 * (18–180)
3. Big Double Creek	19	2.63	0.306	0.315	0.0534 (− 0.13632 to 0.01728)	0.1917	73%	∞ (11–∞)
4. Laurel/Cortland Fork	26	4.63	0.559	0.569	0.0364 (− 0.05947 to 0.09079)	0.5110	91%	132.2 (35–∞)
Middle Fork Kentucky R.								
5. Hell For Certain Creek	24	2.27	0.307	0.275	-0.0929 (− 0.23727 to 0.01583)	1.00	73%	33.9 (6– ∞)
North Fork Kentucky R.								
6. Frozen Creek	27	3.72	0.466	0.462	0.0103 (− 0.08065 to 0.06404)	**0.0061**	91%	21.1 * (12–51)
7. Hell Creek	14	3.72	0.535	0.542	0.0511 (− 0.12061 to 0.12852)	0.0606	91%	96.7 (12–∞)
8. Clemons Fork	31	4.0	0.398	0.433	0.0984 (− 0.02611 to 0.18947)	**0.002****	82%	34.0 * (17–144)
Red R.								
9. Rockbridge Fork	9	1.27	0.020	0.036	0.4921 (− 0.14286 to 0.66667)	0.2249	18%	0.4 (0–∞)

Population identification labels (numbers and stream names) correspond to those in Online Resource 2 and used throughout the document. Populations are grouped by larger river system

Table 3 Pairwise F_{ST} (Weir and Cockerham 1984) values for the nine populations of Kentucky Arrow Darter, *Etheostoma s. spilotum*, examined using microsatellite loci

Populations by river system	1. Wild Dog	2. Upper Bear	3. Big Double	4. Laurel/ Cortland	5. Hell for Certain	6. Frozen	7. Hell	8. Clemons
Sturgeon Cr.								
1. Wild Dog Creek (18)	–							
South Fork Kentucky R.								
2. Upper Bear Creek (24)	0.19	–						
3. Big Double Creek (19)	0.38	0.34	–					
4. Laurel/Cortland Fork (26)	0.14	0.13	0.40	–				
Middle Fork Kentucky R.								
5. Hell for Certain Creek (24)	0.34	0.41	0.63	0.39	–			
North Fork Kentucky R.								
6. Frozen Creek (27)	0.29	0.27	0.47	0.28	0.47	–		
7. Hell Creek (14)	0.19	0.21	0.46	0.20	0.44	0.24	–	
8. Clemons Fork (31)	0.23	0.29	0.55	0.22	0.40	0.27	0.29	–
Red R.								
9. Rockbridge Fork (9)	0.49	0.49	0.74	0.47	0.70	0.52	0.58	0.49

Numbers in parentheses are number of individuals genotyped for a given population. All pairwise comparisons are significant ($p \leq 0.0001$). Population identification labels (numbers and stream names) correspond to those in Online Resource 2 and used throughout the document. Populations are grouped by larger river system

Table 4 Observed heterozygosity (H_o) for the Genus *Etheostoma*, the Kentucky Arrow Darter, *Etheostoma sagitta spilotum*, and several imperiled and federally listed species of darters. The KAD has one of the lower levels of H_o compared to other imperiled darter species examined in contemporary studies

Species	H_o (range)	H_o (average)
Genus *Etheostoma*[a]	Not provided	0.735
Yazoo Darter[b]	Not provided	0.749
Roanoke Logperch (FE)[c]	0.448–0.837	0.699
Watercress Darter (FE)[d]	0.652–0.702	0.671
Okaloosa Darter (FT)[e]	0.000–0.950	0.599
Kentucky Arrow Darter (FT)	0.036–0.630	0.426
Bluemask Darter (FE)[f]	0.224–0.366	0.292
Citico Darter (FE)[g]	0.000–0.857	0.246

FT Federally Threatened, *FE* Federally Endangered

[a]Estimate of average for species of genus *Etheostoma* as reported in Sterling et al. (2012)

[b]*Etheostoma raneyi*, data summarized from Sterling et al. (2012)

[c]*Percina rex*, data summarized from Roberts et al. (2013)

[d]*Ethestoma nuchale*, data summarized from Fluker et al. (2010)

[e]*Etheostoma okaloosa*, data summarized from Austin et al. (2011)

[f]*Etheostoma akatulo*, summarized from Robinson et al. (2013)

[g]*Etheostoma sitikuense*, data summarized from USFWS (2012)

assignments in which sites were grouped for Laurel and Cortland Forks (pop 4; SFK) and in Hell Creek (pop 7; NFK) were confirmed. No evidence of differentiation between Laurel Fork and Cortland Fork (pop 4; SFK) or the two sites in Hell Creek (pop 7; NFK) suggests considerable gene flow among individuals from the two sample localities within each of these designated populations. However, as also indicated by the significant pairwise F_{ST} values, the nine clusters recovered by STRUCTURE had little to no admixture suggesting each locality contains a population that is genetically isolated from others examined (Figs. 1c and 2 b).

River-kilometer distance was significantly and positively correlated with the estimated degree of genetic isolation among populations ($p = 0.0023$, Fig. 3). However, the r^2 value explained only 41% of the observed F_{ST} values. This and the observed outliers in our regression analysis suggest that genetic isolation is not driven exclusively by the degree of geographical distance among populations.

Point estimates of N_e for all nine populations were relatively low, ranging from 0.4 to 132.2, with all but one (Laurel/Cortland FK, pop 4) below 50 (Table 2). Lower bound estimates for all sites were < 40, but several had infinite upper bound estimates that could be a result of a relatively small number of loci used in the model or violating the LD model used to estimate CIs.

Rockbridge Fork (pop 9; Red River) was excluded from our BOTTLENECK analysis due to low sample size. Three of the remaining eight populations showed signs of a recent bottleneck: Upper Bear Creek (pop 2, SFK; $p = 0.053$), Laurel/Cortland Fork (pop 4, SFK; $p = 0.001$), and Hell Creek (pop 7, NFK; $p = 0.002$). We obtained the same results for all mutation models examined, confirming that each of these

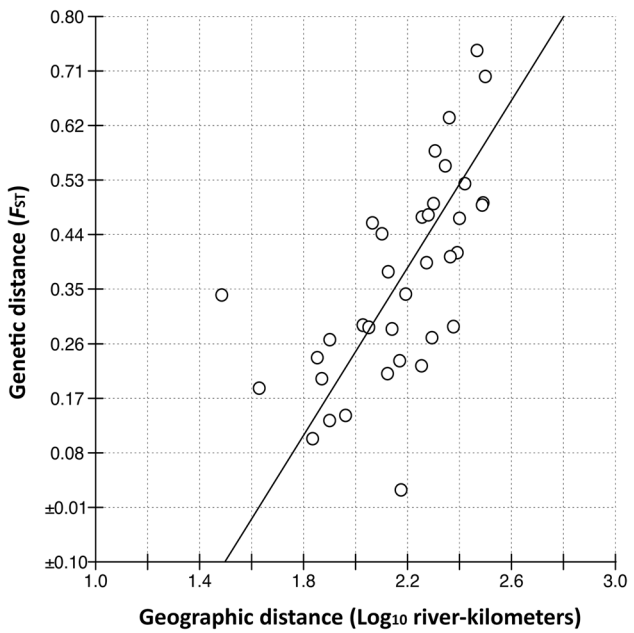

Fig. 3 Isolation by distance regression (RMA, $r^2 = 0.412$, $p = 0.0023$) for the nine populations of Kentucky Arrow Darter, *Etheostoma s. spilotum*, identified from analyses of microsatellite loci, using F_{ST} genetic distances (Table 3) and $\log_{10}$ river kilometer pairwise distances ($Y = -1.13(\pm 0.198) + 0.69(\pm 0.09)x$)

populations has undergone recent, drastic reductions in population size (Williamson-Natesan 2005).

Discussion

The negative influence of habitat fragmentation on all levels of biodiversity is well documented (reviewed in Fahrig 2003), and loss of biodiversity impairs ecosystem function (Lui et al. 2018). Thus, understanding general and species-specific responses to habitat fragmentation has become a primary goal of conservation biology. Using a range-wide survey of genetic diversity in *E. s. spilotum*, we provide another example in which human-mediated habitat fragmentation has contributed to increased population isolation. Our *mt*DNA results do not find evidence for occurrence of long-standing isolation among *E. s. spilotum* populations, supporting the conclusion that isolation has been recent and has contributed to genetic drift, resulting in low genetic diversity within and high genetic differentiation among extant populations.

Contemporary genetic diversity and structure in *Etheostoma sagitta spilotum*

Microsatellite loci revealed low genetic diversity within and a high degree of isolation and genetic structure among *E. s. spilotum* populations. Heterozygosity and allelic richness (H_O average = 0.42, Na = 1.3–5.5) were lower than average diversity values across fishes (H = 0.46, Na = 7.5; DeWoody and Avise 2000) and other *Etheostoma* (Average $H_o = 0.74$; Sterling et al. 2012). Although not directly comparable due to variation in loci used, *E. s. spilotum* populations appear more genetically isolated than those of many other darter species of conservation concern (e.g., Fluker et al. 2010; Austin et al. 2011; Robinson et al. 2013). For example, the Bluemask Darter (*Etheostoma akatulo*), a range-limited species with known isolating events (i.e. extirpation from one river of historic occurrence and presence of the Great Falls Dam), has a higher degree of genetic admixture among populations and less structuring across similar river distances (Robinson et al. 2013) than observed for *E. s. spilotum*.

We expected localities within river systems to comprise single admixed populations, especially those on publicly owned lands that are presumed to provide higher quality corridor conditions due to less disturbance overtime and where population loss has been less severe (USFWS 2013; Hitt et al. 2016). Our data do not support this expectation. Instead, the combined outcomes of significant pairwise F_{ST} values for all localities and recovery of nine distinct genetic clusters with little to no evidence of admixture implies there are at least nine reproductively distinct populations of *E. s. spilotum*. Furthermore, each population has one or more other indicators of diminished evolutionary potential (e.g., small N_e, low H_o, bottlenecks).

A positive, significant isolation by distance relationship is expected in dispersal-limited and obligate freshwater species (Bohanak 1999; Savage et al. 2010). Our data generally conform to this expectation and demonstrate that increased distances between locations contributes to a reduction in genetic exchange and partially explains the observed patterns of genetic distinctiveness among sites examined. For example, some of the highest fixation levels observed involve comparisons with Rockbridge Fork (Red River, pop 9) and Hell for Certain Creek (MFK, pop 5), which are two of the most geographically isolated sites examined. Hell for Certain Creek is further isolated by Buckhorn Lake. However, our data do not entirely fit an isolation-by-distance model. Pairwise comparisons including proximate localities in the SFK system had higher than expected pairwise F_{ST} values (Table 3). These sites occur in the Daniel Boone National Forest and the SFK system is regarded as a stronghold for *E. s. spilotum*. The higher than expected F_{ST} values and lack of admixture among adjacent sites in this system implies factors other than geographic distance have contributed to the observed patterns of genetic structure and isolation, and that presumed quality habitats are not serving as functional corridors for successful dispersal and gene flow (Coulon et al. 2004). Alternatively, currently high-quality habitats could have undergone repeated episodes of degradation over the

past century. *Etheostoma sagitta spilotum* may have been eliminated from these areas long ago, but remain unable to recolonize due to instability caused by repeated physical disturbances and persistent water quality problems.

Genetic variability within fragmented populations will decrease due to genetic drift and inbreeding as a result of decreases in N_e (Young and Clarke 2000; Frankham et al. 2002). All but one of the point estimates of N_e were below 150 and many had upper bound CI estimates lower than 200 individuals, including Upper Bear Creek (pop 2; SFK), Frozen Creek (pop 6; NFK), Clemons Fork (pop 8; NFK), and Rockbridge Fork (pop 9; Red River). The 100/1000 rule states that 100 individuals is the minimum N_e to avoid severe inbreeding, while the minimum N_e necessary to maintain evolutionary potential in a species is 1000 individuals (Frankham et al. 2014). According to this rule, most *E. s. spilotum* populations are at risk for detrimental effects from short-term inbreeding depression and lack sufficient N_e to maintain long-term evolutionary potential (Newman and Pilson 1997). Furthermore, low N_e may be compounded by the low density of adults observed within this species (3–5 individuals/100 m, Thomas 2008 and USFWS 2013; 0.00–0.05 individuals/m^2, Baxter 2015, unpubl. M.S. Thesis). Low density may be inherent and maintained by the territoriality of males during the spawning season and use of specific spawning habitats (Kuehne and Barbour 1983). Therefore, density and N_e may be naturally low in this species and limited by the availability and arrangement of appropriate spawning sites within stream reaches.

Three populations [Upper Bear Creek (pop 2; SFK), Laurel/Cortland Fork (pop 4; SFK), and Hell Creek (pop 7; NFK)] have undergone recent bottleneck events, implying they have experienced recent extreme population decline resulting in heterozygosity excess (Cornuet and Luikart 1996). However, the analysis we used does not effectively test for bottleneck events that occurred as recently as 50 generations ago, or in populations with larger allelic variations (Luikart and Cornuet 1998; Williamson-Natesan 2005). Therefore, it is possible that additional populations have experienced more recent bottleneck events that were not detected by our analysis. The occurrence of fixed alleles within each population supports this assumption. For example, Clemons Fork (pop 8; NFK) and Hell for Certain Creek (pop 5; MFK) both have large sample sizes (31 and 24 respectively) yet exhibit relatively low values for polymorphic loci, revealing allelic loss within these populations. Neither system was significant for bottleneck effects, but the loss of allelic diversity is likely attributable to a past bottleneck event.

Although F_{IS} was somewhat elevated (0.0252) overall, suggesting evidence of inbreeding, the 95% CI overlaps 0. Thus, we cannot confidently assess inbreeding within *E. s. spilotum* populations. This is most likely due to the high number of alleles per locus (mean = 10.18) and relatively low sample sizes (Hartl and Clark 1997) of our dataset. However, it is important to note the low-levels of observed heterozygosity within populations examined, especially for those with larger sample sizes, when compared to other darter species (Table 4). Populations of *E. s. spilotum* have lower levels of heterozygosity than those examined for the federally endangered Watercress Darter (*Etheostoma nuchale),* which has a smaller range that is also fragmented, with populations isolated among only four spring-pool complexes of the upper Black Warrior River, AL (Fluker et al. 2010). Reduced heterozygosity is an early indicator of extinction risk. It suggests loss of beneficial alleles or accumulation of deleterious alleles in a population and the potential for compromised reproductive fitness and reduced evolutionary resilience (Spielman et al. 2004) such as that observed in cases of resistance to novel pathogens due to loss in heterozygosity in Major Histocompatibility Complex (Sommer 2005).

Historical genetic structure in *Etheostoma sagitta spilotum*

Larger mainstem portions of rivers often serve as filters, if not barriers, to movement (and thus gene flow) of small-stream adapted fishes, enhancing opportunities for isolation (Turner and Robison 2006; Krabbenhoft et al. 2008; Beneteau and Mandrak 2009; Hollinsgworth and Near 2009; Lang and Echelle 2011; Sterling et al. 2012). Thus, we considered long-standing genetic structure as a potential inherent property of *E. s. spilotum*. Given the life history characteristics of this small, headwater-adapted species, it is not surprising that the mainstem Kentucky River may have always served as a filter to dispersal, contributing to the low-levels of structure observed among river systems. However, it is important to note that two of the seven ND2 haplotypes recovered were broadly shared within and among several of the major tributaries of the Kentucky River (e.g., the NFK, SFK and Red River); a third was shared among several sites within the NFK. Furthermore, the seven haplotypes only differed by one to four mutations, indicating overall low levels of genetic divergence across the range of the species. These observations imply that the Kentucky River and other mainstem habitats were not hard barriers to dispersal and historically, the species maintained gene flow across its range. A caveat with our current data set, however, is that the patterns observed in the mitochondrial DNA could reflect incomplete lineage sorting among systems or sites and not a lack of isolation among systems or localities.

The pattern of range-wide genetic exchange and historic population expansion, inferred from the mtDNA, contrasts sharply with results from the microsatellite data. This contrast supports a recent reduction or loss of gene flow

among populations. Furthermore, recent population loss and declines in abundance across the species range were attributed to declines in habitat and corridor quality due to previously identified human-mediated threats (Thomas 2008; USFWS 2013, Hitt et al. 2016; see S1). Surveys for *E. s. spilotum* occurrence conducted during 2007–2012 failed to detect the species at nearly 50% of its known historical localities (Thomas 2008; USFWS 2013). Population loss and corridor quality decline has likely created hostile habitat matrix conditions that prevent successful dispersal among populations and would explain the contemporary degree of genetic isolation observed, as well as other hallmarks of genetically impaired populations, such as reduced heterozygosity, evidence of population bottlenecks, and random allele loss.

Implications and recommendations for *Etheostoma sagitta spilotum* conservation

Our results support the following scenario for *E. s. spilotum*: recent population isolation followed by random loss of allelic diversity within and rapid accumulation of genetic divergence between populations. This pattern has been observed in other imperiled darter species, such as the Yazoo Darter (*E. raneyi*, Sterling et al. 2012) and Roanoke Logperch (*P. rex*, Roberts et al. 2013), which have been impacted by increased habitat fragmentation. *Etheostoma sagitta spilotum* has experienced an onslaught of anthropogenic stressors that have decreased habitat quality and population connectivity across its range; the loss of forested watersheds and increased conductivity levels are key factors associated with population loss (Thomas 2008; USFWS 2013; Hitt et al. 2016). Under current conditions, the remaining *E. s. spilotum* populations may have insufficient N_e for long-term persistence and may be lost if conditions persist or worsen (Frankham et al. 2002; Frankham et al. 2014). Actions that improve habitat conditions and particularly corridor quality may prove critical for species persistence. We recommend continued genetic monitoring to establish an adaptive management framework to ensure actions have positive impacts to the genetic diversity of populations and the species overall. For example, reliance on occurrence data and population census size (N_c) as indicators of population and species status alone may mislead managers charged with implementing conservation actions as N_c may be up to five orders of magnitude larger than N_e (Hedgecock, 1994). Population census size has been estimated for three *E. s. spilotum* populations; all were found to have relatively robust N_c estimates. Elisha and Gilberts Big creeks, both tributaries to the Red Bird River system (SFK) located in the Daniel Boone National Forest were estimated to have between 80-1498 individuals (Baxter 2015, unpubl. M.S. thesis). Our estimates of N_e of two nearby creeks of similar size in the Red Bird River system [Upper Bear Creek (pop 2; SFK) and Big Double Creek (pop 3; SFK)] had lower bound N_e values of only 18 and 11 individuals, respectively. In Clemons Fork (our pop 8; NFK), located in Robinson Forest, Baxter (unpubl. data cited in Baxter 2015) estimated a robust N_c of 986 ($\pm$ 1126, 95% CI) individuals compared to our N_e estimate of only 34 for this population. Based on N_c estimates, in part, these populations have been regarded as some of the most robust for *E. s. spilotum*; however, our data imply that they have low genetic diversity or only a small proportion of individuals successfully reproduce in a generation and each shows other signatures of reduced adaptive potential.

We recommend that corridor functionality be examined. Adult *E. s. spilotum* have the potential to move at least 4 km (Baxter 2015, unpubl. M.S. Thesis) and larvae are pelagic for approximately 21–28 days (Shute and Rakes, pers. comm.), suggesting considerable dispersal potential for a darter. However, our data imply habitat matrix conditions are preventing successful migration among populations or habitat patches (Kottelat and Freyhof 2007). This situation is exemplified by populations separated by short river-mile distances and occurring in some of the "most pristine" streams [e.g., those of the Red Bird River system (SFK) in the Daniel Boone National Forest; US Fish and Wildlife Service (USFWS) 2013]. Populations in these areas are functionally isolated, experiencing little to no genetic exchange and have low N_e. Genetic rescue through captive propagation and reintroduction into high quality or restored habitats where populations were extirpated may also enhance overall connectivity and promote potential recolonization and afford more opportunities for genetic exchange. Assisted gene flow (via translocation of individuals) among historically connected populations may also improve genetic diversity (Pavlova et al. 2017). Such actions should increase resiliency to future stochastic events. However, continued genetic monitoring within an adaptive management framework and adherence to guidelines set forth by George et al. (2009) is required to avoid genetic swamping or outbreeding depression (Hendrick 1995; Weeks et al. 2011; Whiteley et al. 2015, Pavlova et al. 2017) and to ensure any conservation actions pursued have the intended outcome of increasing connectivity and genetic diversity for *E. s. spilotum*.

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